Linear Regression

Advanced Statistical Inference

Simone Rossi

Maximum Likelihood Estimation

1. Given a dataset $\mathcal{D} = \{(\boldsymbol{x}_n, y_n)\}_{n=1}^N$ with $\boldsymbol{x}_n \in \mathbb{R}^D$, assume the generative model $y_n = \boldsymbol{w}^\top \boldsymbol{x}_n + \epsilon_n$ where $\epsilon_n \sim \mathcal{N}(0, \sigma^2)$ independently. Write down the log-likelihood $\ell(\boldsymbol{w}) = \log p(\boldsymbol{y} \mid \boldsymbol{w}, \boldsymbol{X})$ and find $\nabla_{\boldsymbol{w}} \ell(\boldsymbol{w}) = 0$ to derive the MLE $\boldsymbol{w}^*$.

2. Let $\hat{\boldsymbol{w}} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y}$ be the least squares estimator. The true data is generated as $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{w}^* + \boldsymbol{\epsilon}$ where $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma^2 \boldsymbol{I})$. Prove that $\mathbb{E}[\hat{\boldsymbol{w}}] = \boldsymbol{w}^*$ by substituting the generative model into the estimator.

3. For a dataset with $\boldsymbol{X} \in \mathbb{R}^{3\times 2}$ and $\boldsymbol{y} \in \mathbb{R}^3$:

$$\boldsymbol{X} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \boldsymbol{y} = \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix}$$

   Compute the ridge regression solution $\boldsymbol{w}^*_\lambda = (\boldsymbol{X}^\top \boldsymbol{X} + \lambda \boldsymbol{I})^{-1} \boldsymbol{X}^\top \boldsymbol{y}$ for $\lambda = 1$ (you can use the Cholesky decomposition to solve the system numerically).

4. Starting from the regularized loss $\mathcal{L}(\boldsymbol{w}) = \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}\|_2^2 + \lambda \|\boldsymbol{w}\|_2^2$, show that this is equivalent to the negative log posterior (up to constants):

$$-\log p(\boldsymbol{w} \mid \boldsymbol{y}, \boldsymbol{X}) = -\log p(\boldsymbol{y} \mid \boldsymbol{w}, \boldsymbol{X}) - \log p(\boldsymbol{w})$$

   with Gaussian likelihood $\mathcal{N}(\boldsymbol{y} \mid \boldsymbol{X}\boldsymbol{w}, \sigma^2 \boldsymbol{I})$ and prior $p(\boldsymbol{w}) = \mathcal{N}(\boldsymbol{w} \mid \boldsymbol{0}, \tau^2 \boldsymbol{I})$. What is λ in terms of σ^2 and τ^2?

5. Starting from the regularized loss *with L1 penalty* $\mathcal{L}(\boldsymbol{w}) = \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}\|_2^2 + \lambda \|\boldsymbol{w}\|_1$, show that this is equivalent to the negative log posterior (up to constants) of a model with Gaussian likelihood and Laplace prior $p(\boldsymbol{w}) \propto \exp(-\beta \|\boldsymbol{w}\|_1)$. What is the relationship between λ and the scale parameter β of the Laplace distribution?

6. The maximum likelihood estimator for σ^2 is $\hat{\sigma}^2_{\mathrm{ML}} = \frac{1}{N}\|\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{w}}_{\mathrm{ML}}\|_2^2$ where $\hat{\boldsymbol{w}}_{\mathrm{ML}} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y}$. Derive this result by maximizing the log-likelihood with respect to σ^2.

Bayesian Linear Regression

1. Given:

   - Prior: $p(\boldsymbol{w}) = \mathcal{N}(\boldsymbol{w} \mid \boldsymbol{0}, \sigma_w^2 \boldsymbol{I})$
   - Likelihood: $p(\boldsymbol{y} \mid \boldsymbol{w}, \boldsymbol{X}) = \mathcal{N}(\boldsymbol{y} \mid \boldsymbol{X}\boldsymbol{w}, \sigma_y^2 \boldsymbol{I})$

   Show that the posterior is Gaussian by computing the exponential of $p(\boldsymbol{y} \mid \boldsymbol{w}, \boldsymbol{X})p(\boldsymbol{w})$ and identifying the posterior precision matrix $\boldsymbol{\Sigma}^{-1} = \frac{1}{\sigma_y^2}\boldsymbol{X}^\top\boldsymbol{X} + \frac{1}{\sigma_w^2}\boldsymbol{I}$ and mean $\boldsymbol{\mu} = \boldsymbol{\Sigma}\left(\frac{1}{\sigma_y^2}\boldsymbol{X}^\top\boldsymbol{y}\right)$. What property do you need to use to show that the posterior is Gaussian?

2. For a new input $\boldsymbol{x}_*$ and posterior $p(\boldsymbol{w} \mid \boldsymbol{y}, \boldsymbol{X}) = \mathcal{N}(\boldsymbol{w} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma})$, the predictive distribution is obtained by marginalizing:

   $$p(y_* \mid \boldsymbol{x}_*, \boldsymbol{y}, \boldsymbol{X}) = \int \mathcal{N}(y_* \mid \boldsymbol{w}^\top \boldsymbol{x}_*, \sigma_y^2)\mathcal{N}(\boldsymbol{w} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma})\,\mathrm{d}\boldsymbol{w}$$

   Show that this equals $\mathcal{N}(y_* \mid \boldsymbol{\mu}^\top \boldsymbol{x}_*, \boldsymbol{x}_*^\top \boldsymbol{\Sigma}\boldsymbol{x}_* + \sigma_y^2)$ using the property that the marginal of a Gaussian is Gaussian and the linear transformation of a Gaussian is Gaussian.

3. You need to predict the price of a house based on its size, location, number of bedrooms, and other features (total of D features). You have a dataset of N houses with their features and prices. Describe how you would use Bayesian linear regression to make predictions for new houses, including how you would choose the prior and how you would compute the posterior predictive distribution.

Model Selection

1. Suppose two models $\boldsymbol{\mathcal{M}}_1$ (linear) and $\boldsymbol{\mathcal{M}}_2$ (polynomial degree 5) are fit to data. Model $\boldsymbol{\mathcal{M}}_2$ always achieves a higher likelihood $p(\boldsymbol{y} \mid \hat{\boldsymbol{w}}_2, \boldsymbol{X}, \boldsymbol{\mathcal{M}}_2) > p(\boldsymbol{y} \mid \hat{\boldsymbol{w}}_1, \boldsymbol{X}, \boldsymbol{\mathcal{M}}_1)$ on the training set. However, Bayesian model selection chooses $\boldsymbol{\mathcal{M}}_1$. Explain why the marginal likelihood $p(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{\mathcal{M}})$ provides a better criterion than the marginal likelihood of the best-fit parameters.

2. Consider two models for the observed data $\boldsymbol{y}$:

   - $\boldsymbol{\mathcal{M}}_1$: Likelihood $p(\boldsymbol{y} \mid \boldsymbol{w}, \boldsymbol{X}, \boldsymbol{\mathcal{M}}_1) = \mathcal{N}(\boldsymbol{y} \mid \boldsymbol{X}\boldsymbol{w}, 0.1^2\boldsymbol{I})$ with prior $p(\boldsymbol{w}) = \mathcal{N}(\boldsymbol{w} \mid 0, \boldsymbol{I})$
   - $\boldsymbol{\mathcal{M}}_2$: Likelihood $p(\boldsymbol{y} \mid \boldsymbol{w}, \boldsymbol{X}, \boldsymbol{\mathcal{M}}_2) = \mathcal{N}(\boldsymbol{y} \mid \boldsymbol{X}\boldsymbol{w}, 1^2\boldsymbol{I})$ with prior $p(\boldsymbol{w}) = \mathcal{N}(\boldsymbol{w} \mid 0, \boldsymbol{I})$

   Which model assigns higher marginal likelihood to a large-variance dataset? (Hint: the marginal likelihood for a Gaussian regression is $p(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{\mathcal{M}}) = \mathcal{N}(\boldsymbol{y} \mid \boldsymbol{0}, \boldsymbol{X}\boldsymbol{\Sigma}_p\boldsymbol{X}^\top + \sigma^2\boldsymbol{I})$ where $\boldsymbol{\Sigma}_p$ is the prior covariance.)